Variational Nonlinear Kalman Filtering With Unknown Process Noise Covariance

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Motivated by the maneuvering target tracking with sensors such as radar and sonar, this article considers the joint and recursive estimation of the dynamic state and the time-varying process noise covariance in nonlinear state-space models. Due to the nonlinearity of the models and the nonconjugate prior, the state estimation problem is generally intractable as it involves integrals of general nonlinear functions and unknown process noise covariance, resulting in the posterior probability distribution functions lacking closed-form solutions. This article presents a recursive solution for joint nonlinear state estimation and model parameters identification based on the approximate Bayesian inference principle. The stochastic search variational inference is adopted to offer a flexible, accurate, and effective approximation of the posterior distributions. We make two contributions compared to existing variational inference-based noise adaptive filtering methods. First, we introduce an auxiliary latent variable to decouple the latent variables of dynamic state and process noise covariance, thereby improving the flexibility of the posterior inference. Second, we split the variational lower bound optimization into conjugate and nonconjugate parts, whereas the conjugate terms are directly optimized that admit a closed-form solution and the nonconjugate terms are optimized by stochastic gradient, achieving the tradeoff between inference speed and accuracy. The performance of the proposed method is verified on radar target tracking applications by both simulated and real-world data.

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I. INTRODUCTION

In most state estimation problems, it is generally required to perform the nonlinear state estimation in the presence of uncertain model parameters. For instance, there exists nonlinear coordinate transformation in maneuvering target tracking with remote sensing systems such as radar and sonar [1], whereas the target dynamics are generally modeled in Cartesian coordinates, and sensor measurements are processed in polar coordinates. Meanwhile, the model parameters, especially the process noise covariance, are uncertain, arising from the unexpected maneuver of the non-cooperative target. Due to the nonlinearity of the models and the uncertainty of the model parameters, the state estimation problem is generally intractable as it involves integrals of the general nonlinear functions and the unknown model parameters, resulting in the posterior probability distribution function (PDF) of the system state lacking closed-form solutions.

The problems of nonlinear state estimation and model parameter identification have their own line of research, which have been active for decades in the target tracking community. Specifically, the nonlinear state estimation techniques can be generally classified into function approximation, moment approximation, and stochastic approximation. Function approximation techniques approximate a stationary nonlinear dynamical system with a nonstationary linear dynamical system, such as the first-order Taylor series expansion of the extended Kalman filter (EKF) [2]. Moment approximation techniques directly approximate the posterior PDF by Gaussian approximation, where the moments of Gaussian distribution are computed using numerical integration, such as unscented transformation of the unscented Kalman filter (KF) [3], spherical cubature integration of the cubature KF [4], and Gauss–Hermite integration of the quadrature KF [5]. Stochastic approximation techniques, such as the particle filter [6], directly approximate the intractable posterior PDF by Monte Carlo integration, i.e., a set of weighted samples called particles. There are three main categories of methods for model identification: input estimator, multiple model estimator, and noise adaptive estimator. The input estimator treats the unknown parameters as extra input variables and estimates them with the system state jointly [7]. Multiple model estimator [8] performs the joint estimation of model probabilities and system state by using a bank of different hypothetical models representing the different modes of system state. The noise adaptive estimator assumes that the statistics (e.g., the mean and covariance) of the noise are nonstationary and unknown, and performs the noise statistics identification and state estimation simultaneously. Traditionally, the linear state estimation with unknown noise statistics is known as adaptive Kalman filtering.

As aforementioned, finding the optimal and analytical solution of nonlinear state estimation in the presence of model parameter uncertainties is intractable. One should resort to approximate inference. The sampling-based approximate inference, including sequential Monte Carlo (SMC), has been widely used in nonlinear state estimation. However, it suffers from an expensive computational burden, especially for high-dimensional state estimation. As an approach of approximate inference, variational Bayes (VB) [9] yields a deterministic approximation procedure that converts an intractable inference of posterior PDF into optimization. That is, VB posits a class of tractable variational distributions family over the latent space and tries to find the closest distribution to the posterior PDF in terms of Kullback–Leibler (KL) divergence via the calculus of variations. VB-based methods have attracted significant attention in nonlinear state estimation and adaptive filtering applications due to their efficient computation compared to sampling-based methods.

In the aspect of nonlinear state estimation, Šmídl and Quinn [10] proposed a VB-based nonlinear state estimation, where VB was used to accelerate the marginalized particle filtering. Frigola et al. [11] addressed the Bayesian learning of nonparametric nonlinear state-space models, where the nonlinear function was modeled by Gaussian processes with parameters being learned by VB. Gultekin and Paisley [12] presented three nonlinear filtering approaches based on three different divergence measures, that is, the forward KL divergence as used in variational inference, the reverse KL divergence as used in expectation-propagation, and the $\alpha-$divergence. Hu et al. [13] proposed a nonlinear filter based on proximal VB approach [14].

In the aspect of noise adaptive filtering, Särkkä and Nummenmaa [15] made the first attempt at VB for joint recursive estimation of dynamic state and the time-varying measurement noise parameters in linear state-space models. By choosing the Gaussian and Inverse-Gamma distributions as conjugate prior distributions of latent variables, the joint posterior distribution of the state and the process noise covariance are updated iteratively based on the mean-field VB method. Subsequently, this work was extended to the state estimation [16] and smoother [17] in the presence of both inaccurate process noise covariance and measurement noise covariance. Different from [16], which regarded the covariance of the predicted state as latent variable, Ma et al. [18] assumed that the conjugate prior distribution directly depends on the underlying latent variables by inducing auxiliary latent variable, resulting in more accurate state estimation. Mbalawata et al. [19] proposed the variational Bayesian adaptive Metropolis algorithm for nonlinear state estimation with unknown measurement noise, whereas the proposal distribution of the Metropolis algorithm is updated by the work of Särkkä and Nummenmaa [15]. By choosing Wishart distribution as the conjugate prior distribution of the unknown information matrix, Dong et al. [20] proposed a VB adaptive cubature information filter for joint state estimation and measurement noise covariance identification in nonlinear state-space models.

However, to our best knowledge, little work has been done on the nonlinear state estimation in the presence of unknown process noise covariance based on the VB approach. There are two main challenges when using the existing mean-field VB approach for this problem. First, it is challenging to design the joint conjugate prior distributions on latent variables of the system state and process noise covariance [21]. Second, due to the nonlinearity, it is intractable to directly maximize the evidence lower bound (ELBO) by the coordinate ascent algorithm. The nonconjugate prior and nonlinearity make the computation of ELBO intractable. Stochastic optimization [22] allows direct optimization of the ELBO by Monte Carlo integration to fit the variational parameters, bridging the gap between VB and Monte Carlo methods [23], [24]. Paisley et al. [25] proposed a stochastic optimization method for sampling unbiased gradients of ELBO and used control variates to reduce the variance of the stochastic gradient.

This article encapsulates the nonlinear state estimation with uncertain process noise covariance into a variational inference problem. The nonconjugate prior and model nonlinearity disable the existing VB-based noise adaptive filtering approaches. We introduce an auxiliary latent variable to decouple the system state and process noise covariance, such that the auxiliary ELBO is more easily optimized than the traditional ELBO, which improves the flexibility of the posterior PDF inference. Meanwhile, we split the ELBO into nonconjugate parts (arising from the nonlinearity of the system model) and conjugate parts, where the nonconjugate parts are optimized by stochastic gradient and conjugate parts are optimized directly, achieving the tradeoff between inference speed and accuracy. The performance of the proposed method is verified on radar target tracking applications by both simulated and real data.

The rest of this article is organized as follows. Section II describes the problem formulation of nonlinear state estimation with unknown process noise covariance. Section III presents the proposed nonlinear adaptive Kalman filtering method based on stochastic search variational inference. Sections IV and V provides the performance comparison by simulated and real data, respectively. Finally, Section VI concludes this article.

II. PROBLEM FORMATION

Consider the following discrete-time nonlinear state-space system:

$$\boldsymbol{x}_k = \boldsymbol{f}_k(\boldsymbol{x}_{k-1}) + \boldsymbol{v}_k \tag{1}$$

$$\boldsymbol{y}_k = \boldsymbol{h}_k(\boldsymbol{x}_k) + \boldsymbol{w}_k \tag{2}$$

where $\boldsymbol{x}_k \in \mathbb{R}^{n_x}$ is the target state, $\boldsymbol{y}_k \in \mathbb{R}^{n_y}$ is the sensor measurement, $\boldsymbol{f}_k(\cdot)$ is the known nonlinear state transition function, $\boldsymbol{h}_k(\cdot)$ is the known nonlinear measurement function, and $\boldsymbol{v}_k \in \mathbb{R}^{n_x}$ and $\boldsymbol{w}_k \in \mathbb{R}^{n_y}$ are independent zero-mean Gaussian process noise vector and measurement noise vector with corresponding covariance matrices $\boldsymbol{Q}_k$ and $\boldsymbol{R}_k$, respectively. Time is indexed by k. The initial state vector $\boldsymbol{x}_0$ is assumed to follow Gaussian distribution with mean $\hat{\boldsymbol{x}}_{0|0}$ and covariance matrix $\boldsymbol{P}_{0|0}$. Moreover, $\boldsymbol{x}_0$, $\boldsymbol{v}_k$, and $\boldsymbol{w}_k$ are assumed to be mutually uncorrelated.

The model parameters of the aforementioned state-space system include $\{\boldsymbol{Q}_k, \boldsymbol{R}_k, \hat{\boldsymbol{x}}_{0|0}, \boldsymbol{P}_{0|0}\}$. The initial state estimate $\hat{\boldsymbol{x}}_{0|0}$ and $\boldsymbol{P}_{0|0}$ are often less important since they will get "washed away" by the data after a few time steps. The nonlinear measurement function $\boldsymbol{h}_k(\cdot)$ is sensor-dependent and typically known in advance. We assume $\boldsymbol{R}_k$ is known in this article. For maneuvering target tracking, there exists target motion uncertainty. A tracker does not have access to an accurate dynamic model of the target being tracked, making it difficult to predict the target's motion accurately. There are mainly three maneuvering motion models: equivalent-noise model, multiple models, and unknown-input model. This article focuses on the equivalent-noise model, whereas the transition function $\boldsymbol{f}_k(\cdot)$ is known for describing typical target trajectories and adapt the unknown and time-varying parameter $\boldsymbol{Q}_k$ to describe the target maneuver.

The goal of joint state estimation and model parameter identification is to compute the joint posterior PDF $p(\boldsymbol{x}_k, \boldsymbol{Q}_k|\boldsymbol{y}_{1:k})$. Formally, the well-known recursive Bayesian filtering solution consists of the following predict-update cycle.

1) *Initialization:* The recursion starts from the prior distribution $p(\boldsymbol{x}_0, \boldsymbol{Q}_0)$.
2) *Prediction:* The one-step-ahead predicted PDF for the joint latent variables $p(\boldsymbol{x}_k, \boldsymbol{Q}_k|\boldsymbol{y}_{1:k-1})$ is given by the Chapman–Kolmogorov equation:

$$p(\boldsymbol{x}_k, \boldsymbol{Q}_k|\boldsymbol{y}_{1:k-1}) = \int p(\boldsymbol{x}_k|\boldsymbol{x}_{k-1}, \boldsymbol{Q}_k)p(\boldsymbol{Q}_k|\boldsymbol{Q}_{k-1}) \times p(\boldsymbol{x}_{k-1}, \boldsymbol{Q}_{k-1}|\boldsymbol{y}_{1:k-1})\mathrm{d}\boldsymbol{x}_{k-1}\mathrm{d}\boldsymbol{Q}_{k-1}. \quad (3)$$

3) *Update:* The aforementioned predicted PDF is updated by measurement $\boldsymbol{y}_k$ using the Bayes rule

$$p(\boldsymbol{x}_k, \boldsymbol{Q}_k|\boldsymbol{y}_k) \propto p(\boldsymbol{y}_k|\boldsymbol{x}_k, \boldsymbol{R}_k)p(\boldsymbol{x}_k, \boldsymbol{Q}_k|\boldsymbol{y}_{1:k-1}). \quad (4)$$

In the case of linear Gaussian with known model parameters, the posterior PDF of state $\boldsymbol{x}_k$ has a closed-form analytical solution via Kalman filtering. For nonlinear functions $\boldsymbol{f}_k(\cdot)$, $\boldsymbol{h}_k(\cdot)$, and unknown process noise covariance $\boldsymbol{Q}_k$, all the possible values of $\boldsymbol{Q}_k$ have to be considered in the prediction step, and the update step using measurements has no closed-form solution due to the intractable integral computation arising from the nonlinearity in $\boldsymbol{f}_k(\cdot)$ and $\boldsymbol{h}_k(\cdot)$. One must resort to approximation to perform Bayesian inference in practice. Two main techniques can be used to approximate the intractable joint posterior distribution of $\boldsymbol{x}_k$ and $\boldsymbol{Q}_k$, including SMC-based sampling approximation and VB-based deterministic approximation. The former tends to be more accurate but has higher computational costs. The latter tends to be faster, especially in the case of high-dimensional latent variables. In the next section, we solve the aforementioned recursion steps using the VB inference method.

III. NONLINEAR NOISE ADAPTIVE KALMAN FILTERING

A. Variational Bayesian Inference

As aforementioned, the difficulty of the nonlinear noise adaptive Kalman filtering arises from solving the intractable joint posterior PDF $p(\boldsymbol{x}_k, \boldsymbol{Q}_k|\boldsymbol{y}_{1:k})$, which consists of the recursive steps in (3) and (4). This intractable inference can be approximated by VB methods. Here, we briefly introduce VB and stochastic search VB (SSVB) [25] for the intractable ELBO optimization.

Let $p(z_k, \boldsymbol{y}_k)$ be the probabilistic models for joint latent variables z_k (e.g., $\boldsymbol{x}_k$ and $\boldsymbol{Q}_k$) and observed variables $\boldsymbol{y}_k$. In Bayesian inference, the key is to compute the posterior PDF $p(z_k|\boldsymbol{y}_k)$ recursively. VB transforms the posterior inference into an optimization problem [9]. That is, VB posits a class of distributions $q(z_k; \boldsymbol{\lambda}_k)$ over latent variables with variational parameters $\boldsymbol{\lambda}_k$, and tries to find the closest distribution to the exact posterior distributions $p(z_k|\boldsymbol{y}_k)$. Closeness is measured by the KL divergence, and the closest distribution is called *variational distribution*. Minimizing the KL divergence from the variational distribution $q(z_k; \boldsymbol{\lambda}_k)$ to the posterior distribution $p(z_k|\boldsymbol{y}_k)$ is equivalent to maximizing the ELBO $\mathcal{B}(\boldsymbol{\lambda}_k)$. Thus, the objective of VB is to solve the following optimization problem:

$$p(z_k|\boldsymbol{y}_k) \approx q(z_k; \boldsymbol{\lambda}_k^*) = \arg\max_{\boldsymbol{\lambda}_k} \underbrace{\mathbb{E}_{q(z_k;\boldsymbol{\lambda}_k)}\left[\log p(z_k, \boldsymbol{y}_k) - \log q(z_k; \boldsymbol{\lambda}_k)\right]}_{\mathcal{B}(\boldsymbol{\lambda}_k)}. \quad (5)$$

The choice of variational family $q(z_k; \boldsymbol{\lambda}_k)$ trades off the fidelity of the posterior approximation with the complexity of variational optimization. The simplest (classic) variational family of distributions is the *mean-field* variational family where the latent variables are mutually independent and governed by their parameters $\boldsymbol{\lambda}_{i,k}$. For conjugate exponential models, the mean-field VB often has the closed-form solution of (5) by the coordinate ascent, which repeatedly cycles through and optimizes with respect to each variational parameter $\boldsymbol{\lambda}_{i,k}$. Due to the nonlinearity existing in the joint distribution $p(z_k, \boldsymbol{y}_k)$, the optimization of variational objectives is sometimes intractable since the expectation $\mathbb{E}_{q(z_k;\boldsymbol{\lambda}_k)} \log p(z_k, \boldsymbol{y}_k)$ do not have a closed-form solution.

SSVB directly optimizes the intractable variational objective function $\mathcal{B}(\boldsymbol{\lambda}_k)$ based on stochastic optimization. Assume the ELBO $\mathcal{B}(\boldsymbol{\lambda}_k)$ can be separated into two parts, $\mathbb{E}[f(z_k)]$ and $h(z_k, \boldsymbol{\lambda}_k)$, where $f(z_k)$ is the function of z_k whose expectation is intractable, and $h(z_k, \boldsymbol{\lambda}_k)$ is the rest term in $\mathcal{B}(\boldsymbol{\lambda}_k)$ whose expectation is tractable with closed-form solution. The first step in solving (5) is to take the gradient of the ELBO $\mathcal{B}(\boldsymbol{\lambda}_k)$ as

$$\nabla\mathcal{B}(\boldsymbol{\lambda}_k) = \nabla_{\boldsymbol{\lambda}_k}\mathbb{E}_{q(z_k;\boldsymbol{\lambda}_k)}[f(z_k)] + \nabla_{\boldsymbol{\lambda}_k} h(z_k, \boldsymbol{\lambda}_k). \quad (6)$$

The goal of SSVB is to make a stochastic approximation of the first gradient $\nabla_{\boldsymbol{\lambda}_k}\mathbb{E}_{q(z_k;\boldsymbol{\lambda}_k)}[f(z_k)]$, while the second gradient is tractable.

Use the identity $\nabla q(z_k;\boldsymbol{\lambda}_k) = q(z_k;\boldsymbol{\lambda}_k)\nabla \log q(z_k;\boldsymbol{\lambda}_k)$. The gradient of the expectation can be rewritten as an expectation with respect to the variational distribution, i.e.,

$$\nabla_{\boldsymbol{\lambda}_k}\mathbb{E}_{q(z_k;\boldsymbol{\lambda}_k)}[f(z_k)] = \mathbb{E}_{q(z_k;\boldsymbol{\lambda}_k)}[f(z_k)\nabla_{\boldsymbol{\lambda}_k}\log q(z_k;\boldsymbol{\lambda}_k)]. \quad (7)$$

With this equation in hand, we can stochastically approximate this expectation using Monte Carlo integration with samples extracted from the variational distribution

$$\nabla_{\boldsymbol{\lambda}_k}\mathbb{E}_{q(z_k;\boldsymbol{\lambda}_k)}[f(z_k)] = \frac{1}{N}\sum_{i=1}^{N} f(z_k^i)\nabla_{\boldsymbol{\lambda}_k}\log q(z_k^i;\boldsymbol{\lambda}_k) \quad (8)$$

where $z_k^i \overset{\text{iid}}{\sim} q(z_k;\boldsymbol{\lambda}_k)$, $i = 1,\ldots,N$, and N is the number of samples.

Thus, the gradient of ELBO $\nabla\mathcal{B}(\boldsymbol{\lambda}_k)$ is now tractable by replacing the intractable term $\nabla_{\boldsymbol{\lambda}_k}\mathbb{E}_{q(z_k;\boldsymbol{\lambda}_k)}[f(z_k)]$ with the unbiased stochastic approximation in (8). Finally, stochastic optimization updates the variational parameters $\boldsymbol{\lambda}_k$ at the nth iteration with

$$\boldsymbol{\lambda}_k^{(n+1)} = \boldsymbol{\lambda}_k^{(n)} + \boldsymbol{\rho}_k^{(n)}\nabla\mathcal{B}(\boldsymbol{\lambda}_k^{(n)}) \quad (9)$$

which can be carried out by the Adam algorithm [26], where the learning rate $\boldsymbol{\rho}_k^{(n)}$ follows the Robbins-Monro conditions that $\sum_{n=1}^{\infty}\boldsymbol{\rho}_k^{(n)} = \infty$ and $\sum_{n=1}^{\infty}(\boldsymbol{\rho}_k^{(n)})^2 < \infty$. By decreasing the learning rate $\boldsymbol{\rho}_k^{(n)}$, the ELBO $\mathcal{B}(\boldsymbol{\lambda}_k)$ converges to a local optimal solution.

The practical issue of the stochastic approximation to optimize the ELBO is that the variance of the gradient approximation [under the Monte Carlo estimation in (8)] can be too large to be useful. In order to decrease the variance, larger N and smaller learning rate $\boldsymbol{\rho}_k$ are needed, leading to slow convergence. Variance reduction methods work by replacing the function whose expectation is approximated via Monte Carlo with another function that has the same expectation but a smaller variance. Toward this end, one can introduce a control variate $g(z_k)$, which approximates $f(z_k)$ well and has a closed-form expectation under $q(z_k)$ [25]. Using $g(z_k)$ and a scalar $\alpha_k \in \mathbb{R}$, the new function $\hat{f}(z_k)$ as follows:

$$\hat{f}(z_k) = f(z_k) - \alpha_k\left(g(z_k) - \mathbb{E}_{q(z_k)}[g(z_k)]\right) \quad (10)$$

has the same expectation as $f(z_k)$ but has a smaller variance. The scalar $\alpha_k = \mathrm{Cov}(f,g)/\mathrm{Var}(g)$ is set to minimize the variance. Therefore, one can replace $f(z_k)$ with $\hat{f}(z_k)$ in (8).

B. Prior Distribution and Auxiliary Latent Variable

Recall that the recursive Bayesian filtering consists of initialization, recursive steps of state prediction and update. The initialization step is to model the prior distribution of latent variables. In our models, the prior distribution of the system state $\boldsymbol{x}_k$, i.e., predicted PDF, is assumed to be Gaussian distribution, i.e.,

$$p(\boldsymbol{x}_k|\boldsymbol{y}_{1:k-1},\boldsymbol{Q}_k) = \mathcal{N}(\boldsymbol{x}_k|\hat{\boldsymbol{x}}_{k|k-1},\boldsymbol{P}_{k|k-1}) \quad (11)$$

where $\hat{\boldsymbol{x}}_{k|k-1}$ and $\boldsymbol{P}_{k|k-1}$ are the predicted state and corresponding covariance at time k.

The commonly used prior for the process noise covariance $\boldsymbol{Q}_k$ is the inverse Wishart distribution [17], [20], which serves as a conjugate prior for the positive definite covariance matrix of a multivariate Gaussian distribution. Employing the inverse Wishart prior ensures that the posterior distribution of $\boldsymbol{Q}_k$ is also an inverse Wishart distribution when given Gaussian distributed data. Moreover, this distribution guarantees that the covariance matrix remains positive definite. Thus, we write the prior as

$$p(\boldsymbol{Q}_k|\boldsymbol{y}_{1:k-1}) = \mathrm{IW}(\boldsymbol{Q}_k|\hat{u}_{k|k-1},\boldsymbol{U}_{k|k-1}) \quad (12)$$

where $\mathrm{IW}(\boldsymbol{P}|\nu,\boldsymbol{A})$ signifies that $\boldsymbol{P}$ follows an inverse Wishart distribution with degrees of freedom parameter ν and positive-definite scale matrix $\boldsymbol{A} \in \mathbb{R}^{p\times p}$ (with $\nu > p+1$). The mean value of $\boldsymbol{P}$ is $\mathbb{E}[\boldsymbol{P}] = \boldsymbol{A}/(\nu - p - 1)$. Meanwhile, the inverse $\boldsymbol{P}^{-1}$ follows Wishart distribution and has the mean $\mathbb{E}[\boldsymbol{P}^{-1}] = \nu\boldsymbol{A}^{-1}$. The initial process noise covariance $\boldsymbol{Q}_0$ is assumed to follow an inverse Wishart distribution $\mathrm{IW}(\boldsymbol{Q}_0|\hat{u}_{0|0},\boldsymbol{U}_{0|0})$ with mean value $\bar{\boldsymbol{Q}}_0$ given by

$$\bar{\boldsymbol{Q}}_0 = \frac{\boldsymbol{U}_{0|0}}{\hat{u}_{0|0} - n_x - 1} \quad (13)$$

where the initial value $\hat{u}_{0|0} = \tau + n_x + 1$ and $\boldsymbol{U}_{0|0} = \tau\bar{\boldsymbol{Q}}_0$ with τ being the tuning parameter.

Combining (11) with (12), the joint prior distribution of $\boldsymbol{x}_k$ and $\boldsymbol{Q}_k$ is given by

$$\begin{aligned} p(\boldsymbol{x}_k,\boldsymbol{Q}_k|\boldsymbol{y}_{1:k-1}) &= \mathcal{N}(\boldsymbol{x}_k|\hat{\boldsymbol{x}}_{k|k-1},\boldsymbol{P}_{k|k-1}) \\ &\quad\times \mathrm{IW}(\boldsymbol{Q}_k|\hat{u}_{k|k-1},\boldsymbol{U}_{k|k-1}). \end{aligned} \quad (14)$$

From (14), it is seen that the latent variables $\boldsymbol{x}_k$ and $\boldsymbol{Q}_k$ are dependent since the covariance of the predicted state $\boldsymbol{P}_{k|k-1}$ is related to $\boldsymbol{Q}_k$. Moreover, the joint prior distribution $p(\boldsymbol{x}_k,\boldsymbol{Q}_k|\boldsymbol{y}_{1:k-1})$ is nonconjugate. The work of [15], [19], and [20] performs joint estimation of state $\boldsymbol{x}_k$ and measurement noise covariance $\boldsymbol{R}_k$ based on the mean-field VB, which is inappropriate for modeling the maneuvering target tracking problem. In order to obtain the conjugate prior on joint $\boldsymbol{x}_k$ and $\boldsymbol{P}_{k|k-1}$, Huang et al. [16] regarded $\boldsymbol{P}_{k|k-1}$ as a latent variable, and updated the joint posterior PDFs using VB, which is tractable but indirect since the unknown parameters are $\boldsymbol{Q}_k$ rather than $\boldsymbol{P}_{k|k-1}$.

One way to improve the flexibility of the VB models is by introducing *auxiliary latent variables*, which can be used to reduce correlation between the original variables [27]. In order to decompose $\boldsymbol{P}_{k|k-1}$ in (14), the continuous auxiliary latent variable $\boldsymbol{m}_k$ is introduced. Then, (11) can be rewritten as

$$\begin{aligned} &p(\boldsymbol{x}_k|\boldsymbol{y}_{1:k-1},\boldsymbol{Q}_k) \\ &= \int p(\boldsymbol{x}_k,\boldsymbol{m}_k|\boldsymbol{y}_{1:k-1},\boldsymbol{Q}_k)\mathrm{d}\boldsymbol{m}_k \\ &= \int p(\boldsymbol{x}_k|\boldsymbol{m}_k,\boldsymbol{Q}_k)p(\boldsymbol{m}_k|\boldsymbol{y}_{1:k-1})\mathrm{d}\boldsymbol{m}_k \\ &= \int \mathcal{N}(\boldsymbol{x}_k|\boldsymbol{m}_k,\boldsymbol{Q}_k)\mathcal{N}(\boldsymbol{m}_k|\hat{\boldsymbol{m}}_{k|k-1},\boldsymbol{\Sigma}_{k|k-1})\mathrm{d}\boldsymbol{m}_k \end{aligned} \quad (15)$$

with

$$
\begin{aligned}
\hat{\boldsymbol{m}}_{k|k-1} &= \mathbb{E}_{p(\boldsymbol{x}_{k-1})}[\boldsymbol{f}_k(\boldsymbol{x}_{k-1})] \\
\boldsymbol{\Sigma}_{k|k-1} &= \mathbb{E}_{p(\boldsymbol{x}_{k-1})}[(\boldsymbol{f}_k(\boldsymbol{x}_{k-1}) - \hat{\boldsymbol{m}}_{k|k-1}) \\
&\qquad \times (\boldsymbol{f}_k(\boldsymbol{x}_{k-1}) - \hat{\boldsymbol{m}}_{k|k-1})^T]. \qquad (16)
\end{aligned}
$$

The auxiliary latent variable $\boldsymbol{m}_k \sim \mathcal{N}(\boldsymbol{m}_k|\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k})$ splits $\boldsymbol{P}_{k|k-1}$ into two parts. Now the process noise covariance $\boldsymbol{Q}_k$ is the covariance of the system state $\boldsymbol{x}_k$, making the conjugate prior distribution with Gaussian and Inverse Wishart possible.

The objective of the VB-based method with auxiliary latent variables $\boldsymbol{m}_k$ is to approximate the intractable joint posterior distribution $p(\boldsymbol{x}_k, \boldsymbol{Q}_k, \boldsymbol{m}_k|\boldsymbol{y}_k)$ with the tractable variational distribution $q(\boldsymbol{x}_k, \boldsymbol{Q}_k, \boldsymbol{m}_k; \boldsymbol{\lambda}_k)$. The KL-divergence with auxiliary latent variable $\boldsymbol{m}_k$ can be expressed as

$$
\begin{aligned}
&D_{\mathrm{KL}}(q(\boldsymbol{x}_k, \boldsymbol{Q}_k, \boldsymbol{m}_k)||p(\boldsymbol{x}_k, \boldsymbol{Q}_k, \boldsymbol{m}_k)) \\
&\quad = D_{\mathrm{KL}}(q(\boldsymbol{x}_k, \boldsymbol{Q}_k)||p(\boldsymbol{x}_k, \boldsymbol{Q}_k)) \\
&\qquad + \mathbb{E}_q(D_{\mathrm{KL}}(q(\boldsymbol{m}_k|\boldsymbol{x}_k, \boldsymbol{Q}_k)||q(\boldsymbol{m}_k|\boldsymbol{x}_k, \boldsymbol{Q}_k)). \qquad (17)
\end{aligned}
$$

From (17), it is seen that the ELBO becomes less tight by augmenting an auxiliary latent variable. However, the auxiliary latent variable can lead to significant improvements on approximating the true posterior PDF since one can access to much more flexible class of variational distribution [23], [27], [28], [29].

The conjugate prior distribution of joint latent variables $\boldsymbol{x}_k$, $\boldsymbol{Q}_k$, and $\boldsymbol{m}_k$ is given by

$$
\begin{aligned}
p(\boldsymbol{x}_k, \boldsymbol{Q}_k, \boldsymbol{m}_k|\boldsymbol{y}_{1:k-1}) &= \mathcal{N}(\boldsymbol{x}_k|\boldsymbol{m}_k, \boldsymbol{Q}_k) \\
&\quad \times \mathrm{IW}(\boldsymbol{Q}_k|\hat{u}_{k|k-1}, \boldsymbol{U}_{k|k-1}) \\
&\quad \times \mathcal{N}(\boldsymbol{m}_k|\hat{\boldsymbol{m}}_{k|k-1}, \boldsymbol{\Sigma}_{k|k-1}). \qquad (18)
\end{aligned}
$$

Next, we will determine the prior parameters $\hat{\boldsymbol{m}}_{k|k-1}$, $\boldsymbol{\Sigma}_{k|k-1}$ and $\hat{u}_{k|k-1}$, $\boldsymbol{U}_{k|k-1}$.

According to the definition by (16) and using the Monte Carlo integration, we have

$$
\begin{aligned}
\hat{\boldsymbol{m}}_{k|k-1} &= \frac{1}{N}\sum_{i=1}^{N} \boldsymbol{f}_k\left(\boldsymbol{x}_{k-1}^i\right) \\
\boldsymbol{\Sigma}_{k|k-1} &= \frac{1}{N}\sum_{i=1}^{N} \left(\boldsymbol{f}_k(\boldsymbol{x}_{k-1}^i) - \hat{\boldsymbol{m}}_{k|k-1}\right) \\
&\quad \times \left(\boldsymbol{f}_k(\boldsymbol{x}_{k-1}^i) - \hat{\boldsymbol{m}}_{k|k-1}\right)^T \qquad (19)
\end{aligned}
$$

with $\boldsymbol{x}_{k-1}^i \overset{\mathrm{iid}}{\sim} p(\boldsymbol{x}_{k-1})$, $i = 1, \ldots, N$ and N being the number of samples.

In the vein of [15] and [16], the prior parameters $\hat{u}_{k|k-1}$, $\boldsymbol{U}_{k|k-1}$ are given by

$$
\begin{aligned}
\hat{u}_{k|k-1} &= \beta(\hat{u}_{k-1|k-1} - n_x - 1) + n_x + 1 \\
\boldsymbol{U}_{k|k-1} &= \beta \boldsymbol{U}_{k-1|k-1} \qquad (20)
\end{aligned}
$$

where $\beta \in (0, 1]$ is the factor for spreading.

C. Approximated Posterior Distribution Update

We use the mean-field variational family where the latent variables are mutually independent, each governed by a distinct factor in the variational density. Thus, the variational distribution $q(\boldsymbol{x}_k, \boldsymbol{Q}_k, \boldsymbol{m}_k; \boldsymbol{\lambda}_k)$ approximates $p(\boldsymbol{x}_k, \boldsymbol{Q}_k, \boldsymbol{m}_k|\boldsymbol{y}_{1:k})$ with a free form factorization, i.e.,

$$
\begin{aligned}
q(\boldsymbol{x}_k, \boldsymbol{Q}_k, \boldsymbol{m}_k; \boldsymbol{\lambda}_k) &= q(\boldsymbol{x}_k; \hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}) \\
&\quad \times q(\boldsymbol{Q}_k; \hat{u}_{k|k}, \boldsymbol{U}_{k|k}) \\
&\quad \times q(\boldsymbol{m}_k; \hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k}) \qquad (21)
\end{aligned}
$$

with $\boldsymbol{\lambda}_k = \{\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}, \hat{u}_{k|k}, \boldsymbol{U}_{k|k}, \hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k}\}$.

The optimal variational parameters $\boldsymbol{\lambda}_k^*$ can be obtained by maximizing the ELBO $\mathcal{B}(\boldsymbol{\lambda}_k)$, i.e.,

$$
\boldsymbol{\lambda}_k^* = \arg\max_{\boldsymbol{\lambda}_k} \mathcal{B}(\boldsymbol{\lambda}_k) \qquad (22)
$$

with

$$
\begin{aligned}
\mathcal{B}(\boldsymbol{\lambda}_k) = &\int q(\boldsymbol{x}_k, \boldsymbol{Q}_k, \boldsymbol{m}_k; \boldsymbol{\lambda}_k) \\
&\times \log \frac{p(\boldsymbol{x}_k, \boldsymbol{Q}_k, \boldsymbol{m}_k, \boldsymbol{y}_k|\boldsymbol{y}_{1:k-1})}{q(\boldsymbol{x}_k, \boldsymbol{Q}_k, \boldsymbol{m}_k; \boldsymbol{\lambda}_k)} \mathrm{d}\boldsymbol{x}_k \mathrm{d}\boldsymbol{Q}_k \mathrm{d}\boldsymbol{m}_k
\end{aligned}
$$

where the log-likelihood $\mathcal{J}_k = \log p(\boldsymbol{y}_k, \boldsymbol{x}_k, \boldsymbol{Q}_k, \boldsymbol{m}_k|\boldsymbol{y}_{1:k-1})$ can be decomposed as

$$
\begin{aligned}
\mathcal{J}_k = &\log \mathcal{N}(\boldsymbol{y}_k|h(\boldsymbol{x}_k), \boldsymbol{R}_k) + \log \mathcal{N}(\boldsymbol{x}_k|\boldsymbol{m}_k, \boldsymbol{Q}_k) \\
&+ \log \mathcal{N}(\boldsymbol{m}_k|\hat{\boldsymbol{m}}_{k|k-1}, \boldsymbol{\Sigma}_{k|k-1}) \\
&+ \log \mathrm{IW}(\boldsymbol{Q}_k|\hat{u}_{k|k-1}, \boldsymbol{U}_{k|k-1}). \qquad (23)
\end{aligned}
$$

Then, the variational ELBO $\mathcal{B}(\boldsymbol{\lambda}_k)$ can be written as

$$
\begin{aligned}
\mathcal{B}(\boldsymbol{\lambda}_k) = &-\frac{1}{2}\mathbb{E}_{q(\boldsymbol{x}_k)}\left[(\boldsymbol{y}_k - \boldsymbol{h}_k(\boldsymbol{x}_k))^T \boldsymbol{R}_k^{-1}(\boldsymbol{y}_k - \boldsymbol{h}_k(\boldsymbol{x}_k))\right] \\
&-\frac{1}{2}\mathbb{E}_{q(\boldsymbol{x}_k),q(\boldsymbol{m}_k),q(\boldsymbol{Q}_k)}\left[(\boldsymbol{x}_k - \boldsymbol{m}_k)^T \boldsymbol{Q}_k^{-1}(\boldsymbol{x}_k - \boldsymbol{m}_k)\right] \\
&-\frac{1}{2}\mathbb{E}_{q(\boldsymbol{m}_k)}\left[(\boldsymbol{m}_k - \hat{\boldsymbol{m}}_{k|k-1})^T \boldsymbol{\Sigma}_{k|k-1}^{-1}(\boldsymbol{m}_k - \hat{\boldsymbol{m}}_{k|k-1})\right] \\
&-\frac{1}{2}(\hat{u}_{k|k-1} + n_x + 2)\mathbb{E}_{q(\boldsymbol{Q}_k)}[\log|\boldsymbol{Q}_k|] \\
&-\frac{1}{2}\mathrm{Tr}(\boldsymbol{U}_{k|k-1}\mathbb{E}_{q(\boldsymbol{Q}_k)}[\boldsymbol{Q}_k^{-1}]) \\
&-\mathbb{E}_{q(\boldsymbol{x}_k)}[\log q(\boldsymbol{x}_k)] - \mathbb{E}_{q(\boldsymbol{m}_k)}[\log q(\boldsymbol{m}_k)] \\
&-\mathbb{E}_{q(\boldsymbol{Q}_k)}[\log q(\boldsymbol{Q}_k)] + \mathrm{const}. \qquad (24)
\end{aligned}
$$

Next, we will derive $q(\boldsymbol{x}_k; \hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$, $q(\boldsymbol{Q}_k; \hat{\boldsymbol{u}}_{k|k}, \boldsymbol{U}_{k|k})$ and $q(\boldsymbol{m}_k; \hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k})$, respectively.

1) *Derivations of* $q(\boldsymbol{x}_k; \hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$*:* Rewrite the ELBO $\mathcal{B}(\boldsymbol{\lambda}_k)$ as the function of state $\boldsymbol{x}_k$ and omit the rest terms that are independent of $\boldsymbol{x}_k$, denoted by $\mathcal{B}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$ as follows: (25) shown at the bottom of the next page. The ELBO $\mathcal{B}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$ in (25) can be divided into two terms, where $\mathcal{D}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$ contains the nonlinear function $\boldsymbol{h}_k(\boldsymbol{x}_k)$ that makes the optimization of gradient intractable, and $\mathcal{E}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$ is tractable. To deal with the intractable terms $\mathcal{D}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$, common approaches typically involve making tractable approximations to the nonlinear function $\boldsymbol{h}_k(\boldsymbol{x}_k)$, and then, compute the expectation. For example, one such approximation would pick a point $\boldsymbol{x}_k^*$ and make the first-order Taylor approximation $\boldsymbol{h}_k(\boldsymbol{x}_k) \approx \boldsymbol{h}_k(\boldsymbol{x}_k^*) + \boldsymbol{H}_k(\boldsymbol{x}_k^*)(\boldsymbol{x}_k - \boldsymbol{x}_k^*)$ with $\boldsymbol{H}_k(\boldsymbol{x}_k^*)$ being the Jacobian matrix of $\boldsymbol{h}_k(\boldsymbol{x}_k)$. One then replaces $\boldsymbol{h}_k(\boldsymbol{x}_k)$ in $\mathcal{D}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$ with the linear approximation and optimizes $\mathcal{B}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$.

As stated in [14], this approximation is not tight and can result in a lousy performance. In this case, the resulting update of $q(\boldsymbol{x}_k)$ is the same as the iterative EKF with unknown model parameters.

Since the gradient of $\mathcal{D}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$ is not available in the closed form, we calculate the gradient $\nabla \mathcal{B}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$ by sampling from the approximate PDF $q(\boldsymbol{x}_k)$. That is,

$$\nabla \mathcal{B}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}) = \nabla \mathcal{D}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}) + \nabla \mathcal{E}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}). \tag{26}$$

Next we calculate $\nabla \mathcal{D}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$ and $\nabla \mathcal{E}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$, respectively. Let $S(\boldsymbol{x}_k) = -\frac{1}{2}(\boldsymbol{y}_k - h(\boldsymbol{x}_k))^T \boldsymbol{R}_k^{-1}(\boldsymbol{y}_k - h(\boldsymbol{x}_k))$, we have

$$\nabla \mathcal{D}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}) = \mathbb{E}_{q(\boldsymbol{x}_k)}\left[S(\boldsymbol{x}_k)\nabla \log q(\boldsymbol{x}_k)\right] \tag{27}$$

where the identity $\nabla q(\boldsymbol{x}_k) = q(\boldsymbol{x}_k)\nabla \log q(\boldsymbol{x}_k)$ is used.

The gradient in (27) can be approximated by Monte Carlo integration with samples from the variational distribution

$$\mathbb{E}_{q(\boldsymbol{x}_k)}\left[S(\boldsymbol{x}_k)\nabla \log q(\boldsymbol{x}_k)\right] \approx \frac{1}{N}\sum_{i=1}^{N} S(\boldsymbol{x}_k^i)\nabla \log q(\boldsymbol{x}_k^i) \tag{28}$$

where $\boldsymbol{x}_k^i \overset{\text{iid}}{\sim} q(\boldsymbol{x}_k)$, $i = 1, \ldots N$ with N being the number of samples. The variance reduction methods, which introduce a *control variates* $G(\boldsymbol{x}_k)$ that is highly correlated with $S(\boldsymbol{x}_k)$ but has an analytic expectation, is employed to reduce the variance of the Monte Carlo integration. Using the control variate $G(\boldsymbol{x}_k)$, the gradient in (27) is rewritten as

$$\begin{aligned}\nabla \mathcal{D}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}) &= \alpha_k \nabla \mathbb{E}_{q(\boldsymbol{x}_k)}[G(\boldsymbol{x}_k)] \\ &\quad + \mathbb{E}_{q(\boldsymbol{x}_k)}\left[(S(\boldsymbol{x}_k) - \alpha_k G(\boldsymbol{x}_k))\nabla \log q(\boldsymbol{x}_k)\right]\end{aligned} \tag{29}$$

where $\alpha_k = \mathrm{Cov}(S, G)/\mathrm{Var}(G)$ is set to minimize the variance.

The design of the control variate $G(\boldsymbol{x}_k)$ should meet two criteria: it should be an approximation of $S(\boldsymbol{x}_k)$, and its expectation $\mathbb{E}_{q(\boldsymbol{x}_k)}[G(\boldsymbol{x}_k)]$ should be tractable. There may exist different control variates $G(\boldsymbol{x}_k)$ that can satisfy these requirements. For simplicity, building on the EKF framework where the nonlinear function $\boldsymbol{h}_k(\boldsymbol{x}_k)$ is approximated by first-order Taylor expansion at the expectation value $\hat{\boldsymbol{x}}_{k|k}$ of $q(\boldsymbol{x}_k)$, i.e., $\boldsymbol{h}_k(\boldsymbol{x}_k) \approx \tilde{\boldsymbol{h}}_k(\hat{\boldsymbol{x}}_{k|k-1}, \boldsymbol{x}_k) \triangleq \boldsymbol{h}_k(\hat{\boldsymbol{x}}_{k|k-1}) + \boldsymbol{H}_k(\hat{\boldsymbol{x}}_{k|k-1})(\boldsymbol{x}_k - \hat{\boldsymbol{x}}_{k|k-1})$, the control variate $G(\boldsymbol{x}_k)$ can be designed as [12]

$$\begin{aligned}G(\boldsymbol{x}_k) = &-\frac{1}{2}\left(\boldsymbol{y}_k - \tilde{\boldsymbol{h}}_k\left(\hat{\boldsymbol{x}}_{k|k-1}, \boldsymbol{x}_k\right)\right)^T \boldsymbol{R}_k^{-1} \\ &\times \left(\boldsymbol{y}_k - \tilde{\boldsymbol{h}}_k\left(\hat{\boldsymbol{x}}_{k|k-1}, \boldsymbol{x}_k\right)\right).\end{aligned} \tag{30}$$

As stated in [12], the EKF algorithm corresponds to optimizing the objective (25) by replacing $S(\boldsymbol{x}_k)$ with $G(\boldsymbol{x}_k)$ through a first-order Taylor approximation. The resulting optimization of this approximation does not guarantee minimizing the KL-divergence. SSVB optimizes the objective (25) using stochastic gradient descent, enabling approximate posterior inference with a smaller KL-divergence than an EKF. Specifically, the intractable gradient of the nonlinear function $S(\boldsymbol{x}_k)$ is approximated using Monte Carlo integration [see (28)], and the control variate $G(\boldsymbol{x}_k)$ is introduced to reduce the variance of (28).

Substituting (30) into (29), gradients $\nabla_{\hat{\boldsymbol{x}}_{k|k}} \mathcal{D}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$ and $\nabla_{\boldsymbol{P}_{k|k}} \mathcal{D}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$ are given as (31) shown at the bottom of this page. The term $\mathcal{E}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$ in (25) can be written as

$$\begin{aligned}\mathcal{E}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}) = &\frac{1}{2}\log|\boldsymbol{P}_{k|k}| - \frac{1}{2}\mathrm{Tr}\left(\hat{u}_{k|k}\boldsymbol{U}_{k|k}^{-1}\boldsymbol{P}_{k|k}\right) \\ &-\frac{1}{2}\hat{u}_{k|k}\left[\left(\hat{\boldsymbol{x}}_{k|k} - \hat{\boldsymbol{m}}_{k|k}\right)^T \boldsymbol{U}_{k|k}^{-1}\left(\hat{\boldsymbol{x}}_{k|k} - \hat{\boldsymbol{m}}_{k|k}\right)\right].\end{aligned} \tag{32}$$

The gradients $\nabla_{\hat{\boldsymbol{x}}_{k|k}} \mathcal{E}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$ and $\nabla_{\boldsymbol{P}_{k|k}} \mathcal{E}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$ are given as

$$\begin{aligned}\nabla_{\hat{\boldsymbol{x}}_{k|k}} \mathcal{E}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}) &= -\hat{u}_{k|k}\boldsymbol{U}_{k|k}^{-1}(\hat{\boldsymbol{x}}_{k|k} - \hat{\boldsymbol{m}}_{k|k}) \\ \nabla_{\boldsymbol{P}_{k|k}} \mathcal{E}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}) &= \frac{1}{2}\boldsymbol{P}_{k|k}^{-1} - \frac{1}{2}\hat{u}_{k|k}\boldsymbol{U}_{k|k}^{-1}.\end{aligned} \tag{33}$$

$$\begin{aligned}\mathcal{B}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}) = &\underbrace{-\frac{1}{2}\mathbb{E}_{q(\boldsymbol{x}_k)}\left[(\boldsymbol{y}_k - \boldsymbol{h}_k(\boldsymbol{x}_k))^T \boldsymbol{R}_k^{-1}(\boldsymbol{y}_k - \boldsymbol{h}_k(\boldsymbol{x}_k))\right]}_{\mathcal{D}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})} \\ &\underbrace{-\frac{1}{2}\mathbb{E}_{q(\boldsymbol{x}_k), q(\boldsymbol{m}_k), q(\boldsymbol{Q}_k)}\left[(\boldsymbol{x}_k - \boldsymbol{m}_k)^T \boldsymbol{Q}_k^{-1}(\boldsymbol{x}_k - \boldsymbol{m}_k)\right] - \mathbb{E}_{q(\boldsymbol{x}_k)}[\log q(\boldsymbol{x}_k)]}_{\mathcal{E}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})}.\end{aligned} \tag{25}$$

$$\begin{aligned}\nabla_{\hat{\boldsymbol{x}}_{k|k}} \mathcal{D}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}) = &\frac{1}{N}\sum_{i=1}^{N}\left(S(\boldsymbol{x}_k^i) - G(\boldsymbol{x}_k^i)\right)\left(\boldsymbol{P}_{k|k}^{-1}\boldsymbol{x}_k^i - \boldsymbol{P}_{k|k}^{-1}\hat{\boldsymbol{x}}_{k|k}\right) \\ &+ \boldsymbol{H}_k(\hat{\boldsymbol{x}}_{k|k-1})^T \boldsymbol{R}_k^{-1}\left(\boldsymbol{y}_k - \boldsymbol{h}_k(\hat{\boldsymbol{x}}_{k|k-1}) - \boldsymbol{H}_k(\hat{\boldsymbol{x}}_{k|k-1})(\hat{\boldsymbol{x}}_{k|k} - \hat{\boldsymbol{x}}_{k|k-1})\right) \\ \nabla_{\boldsymbol{P}_{k|k}} \mathcal{D}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}) = &\frac{1}{2N}\sum_{i=1}^{N}\left(S(\boldsymbol{x}_k^i) - G(\boldsymbol{x}_k^i)\right)\left(\boldsymbol{P}_{k|k}^{-1}(\boldsymbol{x}_k^i - \hat{\boldsymbol{x}}_{k|k})(\boldsymbol{x}_k^i - \hat{\boldsymbol{x}}_{k|k})^T \boldsymbol{P}_{k|k}^{-1} - \boldsymbol{P}_{k|k}^{-1}\right) \\ &-\frac{1}{2}\boldsymbol{H}_k(\hat{\boldsymbol{x}}_{k|k-1})^T \boldsymbol{R}_k^{-1}\boldsymbol{H}_k(\hat{\boldsymbol{x}}_{k|k-1}).\end{aligned} \tag{31}$$

Substituting (31) and (33) into (26), we have (34) shown at the bottom of this page. According to (9), the variational parameters $\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}$ are updated iteratively as

$$\begin{aligned}\hat{\boldsymbol{x}}_{k|k}^{(n+1)} &= \hat{\boldsymbol{x}}_{k|k}^{(n)} + \boldsymbol{\rho}_x^{(n)}\left[\boldsymbol{C}^{(n)}\nabla_{\hat{\boldsymbol{x}}_{k|k}}\mathcal{B}_x\left(\hat{\boldsymbol{x}}_{k|k}^{(n)}, \boldsymbol{P}_{k|k}^{(n)}\right)\right]\\ \boldsymbol{P}_{k|k}^{(n+1)} &= \boldsymbol{P}_{k|k}^{(n)} + \boldsymbol{\rho}_x^{(n)}\left[\boldsymbol{C}^{(n)}\nabla_{\boldsymbol{P}_{k|k}}\mathcal{B}_x\left(\hat{\boldsymbol{x}}_{k|k}^{(n)}, \boldsymbol{P}_{k|k}^{(n)}\right)\boldsymbol{C}^{(n)}\right] \end{aligned} \tag{35}$$

where $\boldsymbol{C}^{(n)}$ is a symmetric positive definite matrix that is used to prevent numerical instability problems. In the vein of [12], we select $\boldsymbol{C}^{(n)}$ to be equal to $\boldsymbol{P}_{k|k}^{(n)}$.

2) *Derivations of* $q(\boldsymbol{m}_k; \hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k})$*:* Rewrite the ELBO $\mathcal{B}(\boldsymbol{\lambda}_k)$ in (24) as the function of $\boldsymbol{m}_k$ and omit the rest terms that are independent of $\boldsymbol{m}_k$, denoted by $\mathcal{B}_m(\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k})$ as follows (36) shown at the bottom of this page. The gradient of ELBO $\mathcal{B}_m(\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k})$ is tractable, which is derived directly as follows. For $\mathcal{D}_m(\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k})$, its gradients $\nabla_{\hat{\boldsymbol{m}}_{k|k}}\mathcal{D}_m(\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k})$ and $\nabla_{\boldsymbol{\Sigma}_{k|k}}\mathcal{D}_m(\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k})$ are

$$\begin{aligned}\nabla_{\hat{\boldsymbol{m}}_{k|k}}\mathcal{D}_m(\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k}) &= -\boldsymbol{\Sigma}_{k|k-1}^{-1}(\hat{\boldsymbol{m}}_{k|k} - \hat{\boldsymbol{m}}_{k|k-1})\\ \nabla_{\boldsymbol{\Sigma}_{k|k}}\mathcal{D}_m(\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k}) &= -\frac{1}{2}\boldsymbol{\Sigma}_{k|k-1}^{-1}.\end{aligned} \tag{37}$$

For $\mathcal{E}_m(\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k})$, its gradients $\nabla_{\hat{\boldsymbol{m}}_{k|k}}\mathcal{E}_m(\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k})$ and $\nabla_{\boldsymbol{\Sigma}_{k|k}}\mathcal{E}_m(\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k})$ are

$$\begin{aligned}\nabla_{\hat{\boldsymbol{m}}_{k|k}}\mathcal{E}_m\left(\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k}\right) &= \hat{u}_{k|k}\boldsymbol{U}_{k|k}^{-1}\left(\hat{\boldsymbol{x}}_{k|k} - \hat{\boldsymbol{m}}_{k|k}\right)\\ \nabla_{\boldsymbol{\Sigma}_{k|k}}\mathcal{E}_m\left(\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k}\right) &= \frac{1}{2}\boldsymbol{\Sigma}_{k|k}^{-1} - \frac{1}{2}\hat{u}_{k|k}\boldsymbol{U}_{k|k}^{-1}.\end{aligned} \tag{38}$$

Let the gradients of ELBO $\mathcal{B}_m(\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k})$ equal to zeros. The optimal variational parameters $\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k}$ are obtained as

$$\begin{aligned}\hat{\boldsymbol{m}}_{k|k} &= \hat{\boldsymbol{m}}_{k|k-1} + \boldsymbol{G}_k(\hat{\boldsymbol{x}}_{k|k} - \hat{\boldsymbol{m}}_{k|k-1})\\ \boldsymbol{\Sigma}_{k|k} &= \boldsymbol{\Sigma}_{k|k-1} - \boldsymbol{G}_k\boldsymbol{\Sigma}_{k|k-1}\end{aligned} \tag{39}$$

with

$$\boldsymbol{G}_k = \boldsymbol{\Sigma}_{k|k-1}\left(\boldsymbol{\Sigma}_{k|k-1} + \boldsymbol{U}_{k|k}/\hat{u}_{k|k}\right)^{-1}. \tag{40}$$

3) *Derivations of* $q(\boldsymbol{Q}_k; \hat{\boldsymbol{u}}_{k|k}, \boldsymbol{U}_{k|k})$*:* Rewrite the ELBO $\mathcal{B}(\boldsymbol{\lambda}_k)$ in (24) as the function of $\boldsymbol{Q}_k$ and omit the rest terms that are independent of $\boldsymbol{Q}_k$, denoted by $\mathcal{B}_Q(\hat{u}_{k|k}, \boldsymbol{U}_{k|k})$ as follows: Note that the ELBO $\mathcal{B}_Q(\hat{u}_{k|k}, \boldsymbol{U}_{k|k})$ can be rewritten as (41) shown at the bottom of the next page.

$$\begin{aligned}\mathcal{B}_Q(\hat{u}_{k|k}, \boldsymbol{U}_{k|k}) = &-\frac{1}{2}(\hat{u}_{k|k-1} + 1 - \hat{u}_{k|k})\mathbb{E}_{q(\boldsymbol{Q}_k)}\log|\boldsymbol{Q}_k|\\ &-\frac{1}{2}\mathrm{Tr}\left((\boldsymbol{U}_{k|k-1} + \boldsymbol{C}_k - \boldsymbol{U}_{k|k})\mathbb{E}_{q(\boldsymbol{Q}_k)}[\boldsymbol{Q}_k^{-1}]\right)\end{aligned} \tag{42}$$

with

$$\begin{aligned}\boldsymbol{C}_k &= \mathbb{E}_{q(\boldsymbol{x}_k), q(\boldsymbol{m}_k)}[(\boldsymbol{x}_k - \boldsymbol{m}_k)(\boldsymbol{x}_k - \boldsymbol{m}_k)^T]\\ &= (\hat{\boldsymbol{x}}_{k|k} - \hat{\boldsymbol{m}}_{k|k})(\hat{\boldsymbol{x}}_{k|k} - \hat{\boldsymbol{m}}_{k|k})^T + \boldsymbol{P}_{k|k} + \boldsymbol{\Sigma}_{k|k}.\end{aligned} \tag{43}$$

It is seen that the gradient of ELBO $\mathcal{B}_Q(\hat{u}_{k|k}, \boldsymbol{U}_{k|k})$ equals to zero when the optimal variational parameters are

$$\begin{aligned}\hat{u}_{k|k} &= \hat{u}_{k|k-1} + 1\\ \boldsymbol{U}_{k|k} &= \boldsymbol{U}_{k|k-1} + \boldsymbol{C}_k.\end{aligned} \tag{44}$$

D. Summary

The proposed nonlinear adaptive Kalman filtering algorithm based on stochastic search variational Bayesian, referred to as Stochastic search variational Bayesian adaptive Kalman filter (SSVBAKF), for a single time step is summarized in Algorithm 1, which is a recursive solution of joint nonlinear state estimation and model parameters identification, consisting of closed-loop iterations among state estimation $\boldsymbol{x}_k$, auxiliary latent variable $\boldsymbol{m}_k$, and process noise covariance $\boldsymbol{Q}_k$ at each time.

IV. RESULTS FOR SIMULATED DATA

In this section, we evaluate the performance of the proposed SSVBAKF algorithm in four different simulated scenarios of radar maneuvering target tracking. The results for real scenarios will be presented in the next section.

$$\begin{aligned}\nabla_{\hat{\boldsymbol{x}}_{k|k}}\mathcal{B}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}) = &\frac{1}{N}\sum_{i=1}^{N}\left(S(\boldsymbol{x}_k^i) - G(\boldsymbol{x}_k^i)\right)\left(\boldsymbol{P}_{k|k}^{-1}\boldsymbol{x}_k^i - \boldsymbol{P}_{k|k}^{-1}\hat{\boldsymbol{x}}_{k|k}\right) - \hat{u}_{k|k}\boldsymbol{U}_{k|k}^{-1}(\hat{\boldsymbol{x}}_{k|k} - \hat{\boldsymbol{m}}_{k|k})\\ &+ \boldsymbol{H}_k(\hat{\boldsymbol{x}}_{k|k-1})^T\boldsymbol{R}_k^{-1}\left(\boldsymbol{y}_k - \boldsymbol{h}_k(\hat{\boldsymbol{x}}_{k|k}) - \boldsymbol{H}_k(\hat{\boldsymbol{x}}_{k|k-1})(\hat{\boldsymbol{x}}_{k|k} - \hat{\boldsymbol{x}}_{k|k-1})\right)\\ \nabla_{\boldsymbol{P}_{k|k}}\mathcal{B}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}) = &\frac{1}{2N}\sum_{i=1}^{N}\left(S(\boldsymbol{x}_k^i) - G(\boldsymbol{x}_k^i)\right)\left(\boldsymbol{P}_{k|k}^{-1}(\boldsymbol{x}_k^i - \hat{\boldsymbol{x}}_{k|k})(\boldsymbol{x}_k^i - \hat{\boldsymbol{x}}_{k|k})^T\boldsymbol{P}_{k|k}^{-1} - \boldsymbol{P}_{k|k}^{-1}\right)\\ &-\frac{1}{2}\boldsymbol{H}_k(\hat{\boldsymbol{x}}_{k|k})^T\boldsymbol{R}_k^{-1}\boldsymbol{H}_k(\hat{\boldsymbol{x}}_{k|k}) + \frac{1}{2}\boldsymbol{P}_{k|k}^{-1} - \frac{1}{2}\hat{u}_{k|k}\boldsymbol{U}_{k|k}^{-1}.\end{aligned} \tag{34}$$

$$\begin{aligned}\mathcal{B}_m(\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k}) = &\underbrace{-\frac{1}{2}\mathbb{E}_{q(\boldsymbol{m}_k)}\left[(\boldsymbol{m}_k - \hat{\boldsymbol{m}}_{k|k-1})^T\boldsymbol{\Sigma}_{k|k-1}^{-1}(\boldsymbol{m}_k - \hat{\boldsymbol{m}}_{k|k-1})\right]}_{\mathcal{D}_m(\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k})}\\ &\underbrace{-\frac{1}{2}\mathbb{E}_{q(\boldsymbol{x}_k), q(\boldsymbol{m}_k), q(\boldsymbol{Q}_k)}\left[(\boldsymbol{x}_k - \boldsymbol{m}_k)^T\boldsymbol{Q}_k^{-1}(\boldsymbol{x}_k - \boldsymbol{m}_k)\right] - \mathbb{E}_{q(\boldsymbol{m}_k)}[\log q(\boldsymbol{m}_k)]}_{\mathcal{E}_m(\hat{\boldsymbol{m}}_{k|k}, \boldsymbol{\Sigma}_{k|k})}.\end{aligned} \tag{36}$$

Algorithm 1: SSVBAKF: Stochastic Search Variational Nonlinear Adaptive Kalman Filtering.

Require: Measurement $\boldsymbol{y}_k$, approximated posterior PDFs of last time $q(\boldsymbol{x}_{k-1})$, $q(\boldsymbol{m}_{k-1})$, $q(\boldsymbol{Q}_{k-1})$, sample size N, and the maximum number of iterations $I_{\max}$;

Ensure: Approximated posterior PDFs of current time $q(\boldsymbol{x}_k)$, $q(\boldsymbol{m}_k)$, $q(\boldsymbol{Q}_k)$;

1: **Time prediction:**

2: Calculate prior $p(\boldsymbol{m}_k|\boldsymbol{y}_{1:k-1}) = \mathcal{N}(\boldsymbol{m}|\hat{\boldsymbol{m}}_{k|k-1}, \boldsymbol{\Sigma}_{k|k-1})$ via (19).

3: Calculate prior $p(\boldsymbol{Q}_k|\boldsymbol{y}_{1:k-1}) = \mathrm{IW}(\boldsymbol{Q}_k|\hat{u}_{k|k-1}, \boldsymbol{U}_{k|k-1})$ via (20).

4: **Measurement update:**

5: *Initialization:* $q^{(0)}(\boldsymbol{x}_k) = p(\boldsymbol{x}_k|\boldsymbol{y}_{1:k-1})$, $q^{(0)}(\boldsymbol{m}_k) = p(\boldsymbol{m}_k|\boldsymbol{y}_{1:k-1})$, $q^{(0)}(\boldsymbol{Q}_k) = p(\boldsymbol{Q}_k|\boldsymbol{y}_{1:k-1})$

6: **for** each iteration $n = 1 : I_{\max}$ **do**

7: *Update posterior $q(\boldsymbol{x}_k)$:* Sample $\boldsymbol{x}_k^i \overset{\text{iid}}{\sim} q^{(n-1)}(\boldsymbol{x}_k)$ for $i = 1, \ldots, N$, compute the gradients $\nabla_{\hat{\boldsymbol{x}}_{k|k}} \mathcal{B}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$ and $\nabla_{\boldsymbol{P}_{k|k}} \mathcal{B}_x(\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k})$ as in (34), and update variational parameters $\hat{\boldsymbol{x}}_{k|k}$ and $\boldsymbol{P}_{k|k}$ via (35).

8: *Update posterior $q(\boldsymbol{m}_k)$:* update variational parameters $\hat{\boldsymbol{m}}_{k|k}$ and $\boldsymbol{\Sigma}_{k|k}$ via (39).

9: *Update posterior $q(\boldsymbol{Q}_k)$:* update variational parameters $\hat{u}_{k|k}$ and $\boldsymbol{U}_{k|k}$ via (44).

10: **end for**

A. Simulation Scenarios

The target state $\boldsymbol{x}_k = [x_k, \dot{x}_k, y_k, \dot{y}_k]^T$ consists of target position $[x_k, y_k]^T$ and target velocity $[\dot{x}_k, \dot{y}_k]^T$. Given the initial target state $\boldsymbol{x}_0$, the ground truth of the simulated target trajectory is generated by sampling from the Gaussian distribution recursively, i.e., $\boldsymbol{x}_k \sim N(\boldsymbol{F}_k \boldsymbol{x}_{k-1}, \boldsymbol{Q}_k)$. The transition matrix $\boldsymbol{F}_k$ is assumed to be nearly constant velocity model or coordinated turn model, and the process noise covariance $\boldsymbol{Q}_k$ is given by

$$\boldsymbol{Q}_k = \delta_k^2 \boldsymbol{I}_2 \otimes \begin{bmatrix} T^3/3 & T^2/2 \\ T^2/2 & T \end{bmatrix}$$

where $\boldsymbol{I}_2$ is 2×2 identity matrix, δ_k^2 is process noise level, and T is the sampling period.

We consider two types of target maneuvers. The first type, denoted as $\boldsymbol{M_1}$, involves using the nearly constant velocity model to describe typical target trajectories. The target maneuver is represented by varying the process noise level δ_k^2 at different stages. The total number of simulated steps is 300. Specifically, the actual process noise level δ_k^2 is set to δ^2 for $k \in [1, 100)$, $10 \times \delta^2$ for $k \in [100, 200)$, and δ^2 again for $k \in [200, 300]$. The target maneuvers during the middle stages. The second type of target maneuver, denoted as $\boldsymbol{M_2}$, involves using the coordinated turn model to describe target abrupt maneuvers [30]. The total number of simulated steps is 400. The turn rate parameter ω_k is set to $\{0, 3, 0, 5.6, 0, 8.5, 0, -7.25, 0, 7.25\}^\circ/s$ with changes occurring at times $k = \{60, 120, 150, 214, 240, 272, 300, 338, 360\}$, respectively.

The target is detected by a 2-D radar (located at the origin [0m, 0m]), which provides measurements $\boldsymbol{y}_k = [r_k, \theta_k]^T$ consisting of range r_k and azimuth θ_k. The nonlinear measurement function $\boldsymbol{h}_k(\cdot)$ is given by

$$r_k = \sqrt{x_k^2 + y_k^2}, \qquad \theta_k = \arctan\left(\frac{y_k}{x_k}\right).$$

Four different simulated scenarios for maneuvering target tracking are considered as follows.

1) *S1*: *Aerial target tracking with maneuver type* $\boldsymbol{M_1}$.
2) *S2*: *Vehicle target tracking with maneuver type* $\boldsymbol{M_1}$.
3) *S3*: *Ship target tracking with maneuver type* $\boldsymbol{M_1}$.
4) *S4*: *Aerial target tracking with maneuver type* $\boldsymbol{M_2}$.

The corresponding scenario parameters, including the initial state $\boldsymbol{x}_0$, process noise level δ^2, measurement noise covariance $\boldsymbol{R}$, and sampling period T are given in Table I .

The compared methods, including model adaptive filters and noise parameter adaptive filters, are described as follows.

1) Interacting multiple model unscented Kalman filter (IMM-UKF) [8]: This is a model adaptive filter that uses multiple process noise levels $\delta_i^2 \in \{0.1\delta^2, 0.5\delta^2, 1\delta^2, 5\delta^2, 10\delta^2\}$ (five models in total) for state prediction step. The nonlinear measurement update step is carried out by UKF. The mode transition probabilities from mode i to mode j are set to $P_{ij} = 0.8$ for $i = j$ and $P_{ij} = 0.05$ for $i \neq j$.
2) Variational Bayesian adaptive cubature information filter (VBACIF) [20]: This is a VB-based adaptive nonlinear filter with unknown measurement noise covariance, where the integration of nonlinear recursive Bayesian estimation is approximated by a cubature integration rule. The relevant parameters are set to $v_0 = 1$, $V_0 = 2\boldsymbol{R}^{-1}$, tuning parameter $\tau = 3$, and variance decreasing factor $\beta = 1 - e^{-4}$.
3) Variational Bayesian adaptive Kalman filter with measurement conversion (VBAKF-CM): VBAKF [16] is a VB-based adaptive linear filter designed to handle both unknown process and

$$\mathcal{B}_Q(\hat{u}_{k|k}, \boldsymbol{U}_{k|k}) = -\frac{1}{2}(\hat{u}_{k|k-1} + n_k + 2)\mathbb{E}_{q(\boldsymbol{Q}_k)} \log|\boldsymbol{Q}_k| - \frac{1}{2}\mathrm{Tr}(\boldsymbol{U}_{k|k-1}\mathbb{E}_{q(\boldsymbol{Q}_k)}[\boldsymbol{Q}_k^{-1}])] - \frac{1}{2}\mathbb{E}_{q(\boldsymbol{x}_k),q(\boldsymbol{m}_k),q(\boldsymbol{Q}_k)}\left[(\boldsymbol{x}_k - \boldsymbol{m}_k)^T \boldsymbol{Q}_k^{-1}(\boldsymbol{x}_k - \boldsymbol{m}_k)\right] - \mathbb{E}_{q(\boldsymbol{Q}_k)}[\log q(\boldsymbol{Q}_k)]. \quad (41)$$

TABLE I
Scenario Parameters

Scenario	$\boldsymbol{x}_0$	δ^2	$\boldsymbol{R}$	T
S1	$[5e5\text{m}, -100\text{m/s}, 5e5\text{m}, -100\text{m/s}]^T$	$10\text{m}^2/\text{s}^3$	$\text{diag}(1e4\text{m}^2, 5e{-}5\text{rad}^2)$	10s
S2	$[1e4\text{m}, -10\text{m/s}, 1e4\text{m}, -10\text{m/s}]^T$	$2.5\text{m}^2/\text{s}^3$	$\text{diag}(1e2\text{m}^2, 5e{-}6\text{rad}^2)$	2s
S3	$[5e3\text{m}, -5\text{m/s}, 5e3\text{m}, -5\text{m/s}]^T$	$1\text{m}^2/\text{s}^3$	$\text{diag}(25\text{m}^2,\ 1e{-}6\text{rad}^2)$	3s
S4	$[3e4\text{m}, 300\text{m/s}, 3e4\text{m}, 0\text{m/s}]^T$	$10\text{m}^2/\text{s}^3$	$\text{diag}(2.5e3\text{m}^2, 3e{-}4\text{rad}^2)$	1s

TABLE II
ARMSE in Position for Simulated scenarios (m)

Scenario	VBACIF	VBAKF-CM	IMM-UKF	SSVBAKF
S1	1833.13	2094.51	1660.62	**1357.77**
S2	16.92	17.08	15.06	**13.51**
S3	6.31	6.26	6.00	**5.73**
S4	701.90	599.14	597.05	**546.34**

The bold values represent the best estimation performance, characterized by the lowest RMSE values.

TABLE III
Single Step Run Time for Simulated scenarios (ms)

Scenario	VBACIF	VBAKF-CM	IMM-UKF	SSVBAKF
S1	0.12	0.08	0.82	41.59
S2	0.10	0.07	0.81	31.01
S3	0.15	0.09	0.74	3.48
S4	0.16	0.13	0.93	101.78

measurement noise covariances. VBAKF-CM extends VBAKF to nonlinear scenarios by incorporating the unbiased converted measurement operator. In the simulation, the measurement noise covariance is assumed to be known. The initial hyperparameters are set as $u_0 = 7$ and $\boldsymbol{U}_0 = 3\boldsymbol{Q}_0$, while the tuning parameter is $\tau = 3$, and variance decreasing factor is $\beta = 1 - e^{-4}$.

4) SSVBAKF: This is a process noise covariance adaptive nonlinear filter that approximates the integration of nonlinear recursive Bayesian estimation by stochastic search VB. The initial hyperparameters are set as $u_0 = 7$ and $\boldsymbol{U}_0 = 3\boldsymbol{Q}_0$, while the tuning parameter is $\tau = 3$, the variance decreasing factor is $\beta = 1 - e^{-4}$, and the sampling particle number is $N = 1000$.

For VBACIF, VBAKF-CM, and SSVBAKF, the iterative loop will terminate if the difference between the state estimate in two consecutive iterations is less than a given threshold, specifically, $||\hat{\boldsymbol{x}}_k^{(n+1)} - \hat{\boldsymbol{x}}_k^{(n)}||/||\hat{\boldsymbol{x}}_k^{(n+1)}|| < 1e - 7$, or the maximum number of iterations $I_{\max} = 1000$ is reached.

B. Results

To evaluate the target tracking performance, 100 Monte Carlo runs are carried out for each scenario using a computer equipped with an AMD Ryzen 76800H CPU operating at 3.20 GHz. The tracking performance is evaluated using the root mean square error (RMSE) and averaged RMSE (ARMSE) in target position. As shown in Fig. 1 and Table II, SSVBAKF outperforms existing adaptive filters in all scenarios, exhibiting smaller RMSE and ARMSE values. In Scenario *S1*, *S2*, and *S3*, all methods experience a decrease in estimation accuracy when the target maneuvers during the middle $k \in [100, 200)$. In Scenario *S4*, SSVBAKF proves to be more robust compared to the other methods. The reason for this is that the accuracy of the IMM-UKF estimation relies on the assumption that the complete model set is available. The other three methods, VBACIF, VBAKF-CM, and SSVBAKF, all use variational optimization. However, SSVBAKF directly optimizes the joint nonlinear state estimation and unknown process noise covariance, which may contribute to its superior performance compared to VBACIF and VBAKF-CM, where adjustments are made indirectly to the measurement noise covariance and state predict covariance, respectively.

The computational costs of different methods are compared in Table III, with VBAKF-CM showing the lowest computational cost, while SSVBAKF exhibits the highest. The computational cost of IMM-UKF primarily depends on the number of models, whereas the VB-based methods, including VBACIF, VBAKF-CM, and SSVBAKF, are influenced by the number of iterations. The VBACIF and VBAKF-CM update the variational parameters analytically, leading to faster results compared to the gradient-based SSVBAKF algorithm. SSVBAKF approximates the nonlinear posterior state at each iteration using sampling, which is more time consuming. Furthermore, it is evident that as the target maneuverability decreases in different scenarios, the SSVBAKF algorithm demonstrates a significant decrease in runtime: *S4* (101.78 ms) > *S1* (41.59 ms) > *S2* (31.01 ms) > *S3* (3.48 ms). This is attributed to the fact that the calculation of ELBO in the execution of SSVBAKF requires more iterations to achieve convergence when handling stronger target maneuvers. Specifically, the average number of iterations for SSVBAKF in Scenario *S4*, *S1*, *S2*, and *S3* are 389, 159, 117, and 11, respectively. In contrast, the average number of VBACIF and VBAKF-CM iterations for all scenarios are 4 and 5, respectively. In summary, SSVBAKF achieves improved estimation accuracy but requires more computational time compared to other methods.

The ELBO curves of SSVBAKF for different iterations are shown in Fig. 2. At each iterative cycle, the ELBO increases with the number of iterations increase, indicating that the iteration procedure in SSVBAKF converges.

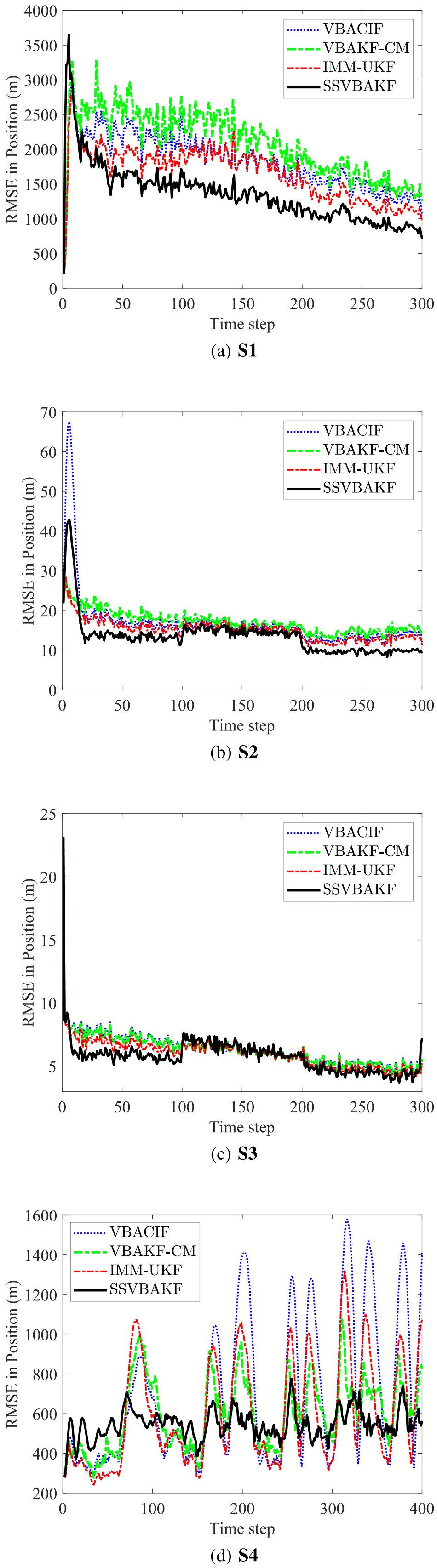


Fig. 1. RMSE comparison in position for simulated scenarios. (a) S1. (b) S2. (c) S3. (d) S4.

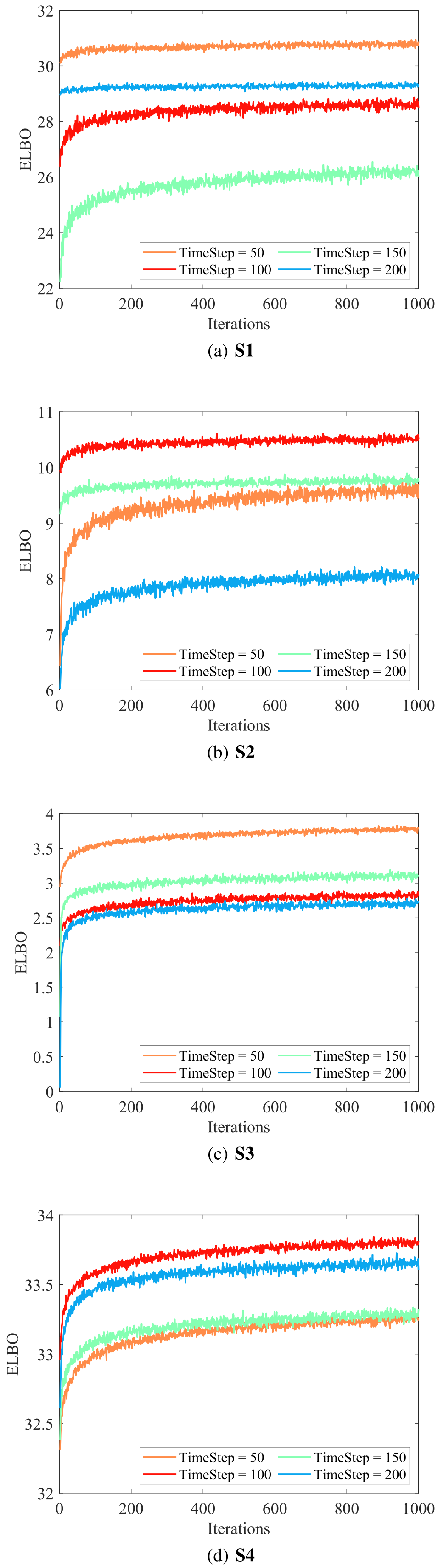


Fig. 2. ELBO curves for simulated scenarios. (a) S1. (b) S2. (c) S3. (d) S4.

V. RESULTS FOR REAL DATA

Next, we apply SSVBAKF to real scenarios, including aerial target tracking, vehicle target tracking, and ship target tracking with different types of 2-D surveillance radar. GPS provides the ground truth of the target trajectory.

A. Real Scenarios

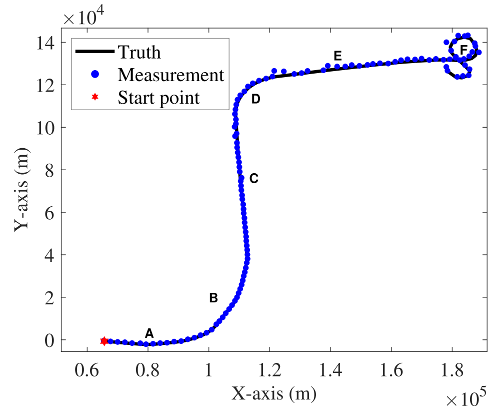
The target trajectory and its corresponding radar detection characteristics for different real scenarios are as follows.

1) *R1*: *Aerial target tracking with composite maneuver*. The trajectory of the aerial target is shown in Fig. 3(a), consisting of six segments (maneuver modes): CV (segment A), left CT (segment B), CV (segment C), right CT (segment D), CA (segment E), and figure-eight flight pattern (segment F). The target is detected by a surveillance radar with scanning period being $T = 10$s, standard deviation of range error being 100 m, and standard deviation of azimuth error being 3.5e-3 rad.
2) *R2*: *Vehicle target tracking with round-trip maneuver*. The trajectory of the vehicle target is shown in Fig. 3(b), which makes a round-trip maneuver. The target is detected by a surveillance radar with scanning period being $T = 2\,s$, standard deviation of range error being 10 m, and standard deviation of azimuth error being 2.5e-3 rad.
3) *R3*: *Ship target tracking with coordinated turn maneuver*. The trajectory of the ship target is shown in Fig. 3(c), which makes turning maneuvers with time-varying turning rates. The target is detected by a surveillance radar with scanning period being $T = 3\,$s, standard deviation of range error being 5 m, and standard deviation of azimuth error being 1e-3 rad.

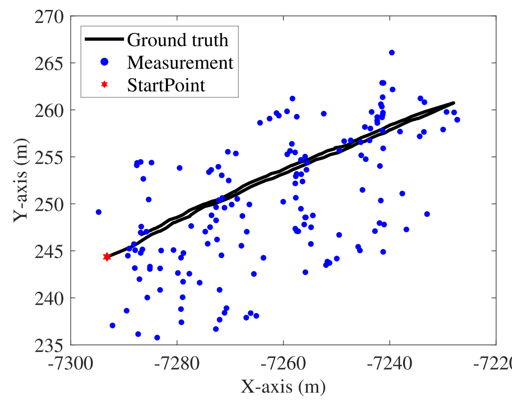
The parameter settings for all algorithms are the same as in simulated scenarios.

B. Results

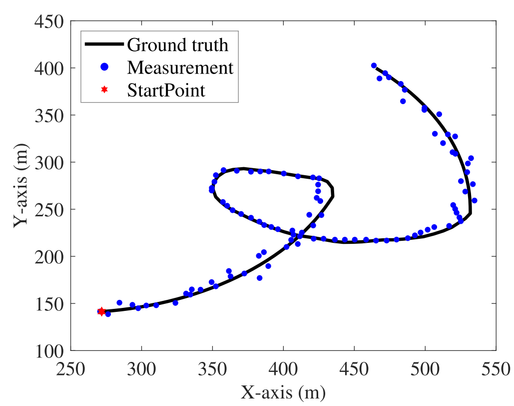
Fig. 4 presents the position RMSE curve for three different real scenarios, and the results align with the simulation scenario, showcasing SSVBAKF's superior filtering performance. In Fig. 4(a), we conduct a segment analysis of the RMSE for *R1*. The result indicates minimal differences in the RMSE curves among the methods in the nonmaneuvering and weakly maneuvering segments. However, SSVBAKF exhibits a more noticeable advantage in the maneuvering segments. Table V records the RMSE comparison for the aerial target in various motion modes. The results reveal that all four methods perform well on filtering when the target moves in a straight line at approximately uniform speed or when the turning maneuver is small (segments A, B, and C). However, when the turning maneuver is significant (segments D and F) or the target is



(a) **R1**



(b) **R2**



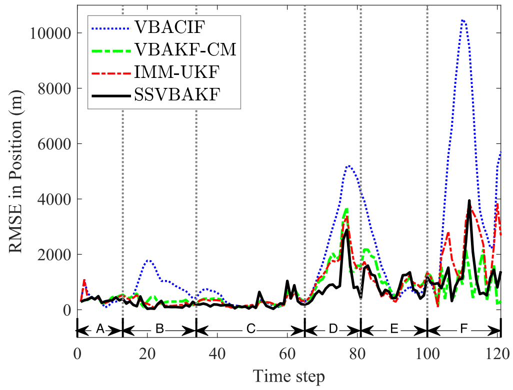
(c) **R3**

Fig. 3. Target trajectories and radar detection for real scenarios. (a) R1. (b) R2. (c) R3.

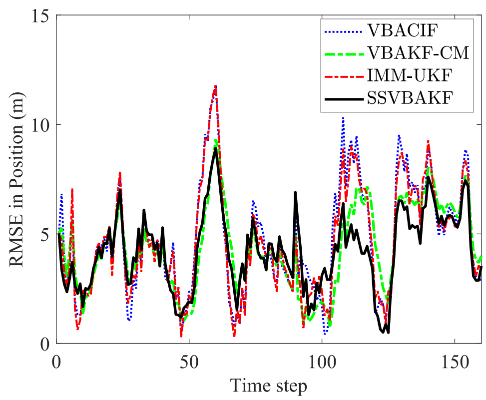
TABLE IV
ARMSE Comparison for Real Scenarios

Scenario	VBACIF	VBAKF-CM	IMM-UKF	SSVBAKF
R1	1784.16	733.39	882.16	**612.28**
R2	4.72	4.38	4.63	**4.21**
R3	5.80	5.96	5.56	**4.61**

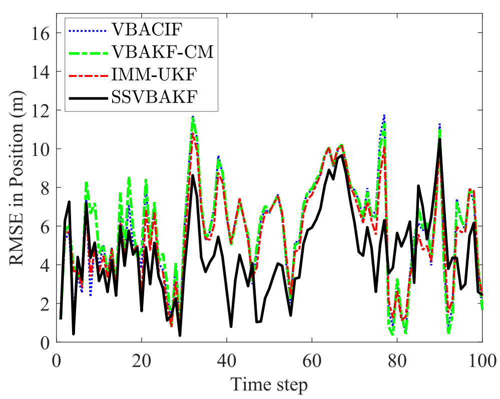
The bold values represent the best estimation performance, characterized by the lowest RMSE values.



(a) R1



(b) R2



(c) R3

Fig. 4. RMSE comparison in position for real scenarios. (a) R1. (b) R2. (c) R3.

TABLE V
Segment ARMSE Comparison for Scenario *R1*

Segment	VBACIF	VBAKF-CM	IMM-UKF	SSVBAKF
A : $[0, 13)$	381.92	386.41	432.63	**363.03**
B : $[13, 34)$	939.07	324.29	235.31	**164.93**
C : $[34, 65)$	352.00	277.76	258.77	**232.87**
D : $[65, 81)$	3051.11	1599.58	1639.06	**977.42**
E : $[81, 100)$	1350.45	1160.58	**902.77**	910.00
F : $[100, 121)$	5038.57	**983.43**	2132.18	1226.44

The bold values represent the best estimation performance, characterized by the lowest RMSE values.

TABLE VI
Single Step Run Time for Real scenarios (ms)

Scenario	VBACIF	VBAKF-CM	IMM-UKF	SSVBAKF
R1	0.25	0.16	0.56	196.50
R2	0.19	0.11	0.26	1.95
R3	0.25	0.14	0.19	158.01

accelerated (segment E), all methods' filtering performance decreases. Nonetheless, SSVBAKF and VBAKF-CM exhibit more robustness to target maneuvers and achieve better performance compared to IMM-UKF and VBACIF. This is because the ability of IMM-UKF to handle target maneuvers relies on the complete and available "model dictionary," which can be challenging to design in practice. Additionally, IMM-UKF faces a model switching delay issue, leading to significant estimation errors in the case of strong maneuvers. VBACIF's ability to handle strong maneuvers is hindered by its indirect optimization mechanism.

Fig. 4(b) and (c) presents the RMSE curves for *R2* and *R3*, respectively. Unlike the aerial target scenario (*R1*), the targets in *R2* and *R3* move in a more homogeneous pattern (straight lines or turns) and move more slowly (especially the ship target in *R3*). Therefore, all four filtering methods can achieve good filtering results. The ARMSE on filtering is given in Table IV. The results show that SSVBAKF has achieved the best filtering results in all scenarios. When the model set of IMM-UKF cannot sufficiently capture the motion mode of the target, the filtering accuracy will deteriorate. In contrast, SSVBAKF estimates the process noise covariance without relying on the initial value of process noise level or the model set. This adaptive estimation ensures that the filtering gain stays within a reasonable range by changing the process noise covariance when the target undergoes maneuvering.

The computational cost comparison of different real scenarios is presented in Table VI, indicating that VBAKF-CM has the lowest computational cost, while SSVBAKF has the highest computational cost. For VBACIF, the average number of iterations in Scenarios *R1*, *R2*, and *R3* are 7, 3, and 5, respectively. For VBAKF-CM, the average number of iterations in Scenarios *R1*, *R2*, and *R3* are 12, 3, and 6, respectively. As for SSVBAKF, the average number of iterations in Scenarios *R1*, *R2*, and *R3* are 750, 6, and 600, respectively. Notably, the average number of iterations for Scenario *R2* is significantly lower than the other two scenarios, primarily due to the relatively weak maneuvering of the vehicle target, allowing the algorithm to converge quickly.

VI. CONCLUSION

In this article, we have proposed an optimization-based nonlinear adaptive Kalman filtering approach to the problem of maneuvering target tracking, by encapsulating the nonlinear state estimation with uncertain process noise covariance into a variational inference problem. To deal with the problems of nonlinearity and the nonconjugate prior encountered by the mean-field variational inference, the stochastic search variational inference with an auxiliary latent variable is adopted to offer a flexible, accurate, and effective approximation of the joint posterior distribution. The proposed method iteratively estimates the process noise covariance and the target state. The simulated and real experiments demonstrated that the proposed method achieved high accuracy and outstanding robustness in different scenarios.

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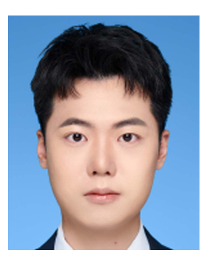

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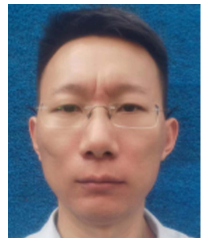

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